Project

Case Study : 1

Data Protection and Cloud Security

Q1) A health care company stores sensitive data of Patients in cloud. Analyse how data anonymisation, tokenization, DLP can be used to protect data. Identify Possible risk & Solution.

Ans) • **Anonymization →**

Irreversibly strips identifiers. Best for research/analytics where individual identity is not needed.

• **Tokenization →**

Replaces sensitive data (SSNs) with a "token" Best for internal operations where systems need to link records without seeing the raw data.

• **DLP →**

Scans for medical codes or Private info in real-time ~~Best~~ Best for Preventing accidental leaks or unauthorized sharing.

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Risk	Solution
• Accidental leaks	DLP Policies to block unauthorized emails.
• ~~Lateral~~ Lateral Movement.	zero trust access & strict encryption at rest.
• Reidentification	Differential Privacy to add noise to datasets.

Case Study - 2

Network and Infrastructure Security

2) A cloud based organization faces frequent network attacks, design a secure architecture using SDN, Vxlan, IDS.

Ans) **SDN (Software Defined Networking) :-**

SDN seperates the control Plane & data Plane, allowing centralized management of the network.

Benefits :-

- easy traffic monitoring
- Fast response to attacks.
- Dynamic Security Policy updates
- Better visibility of Network traffic

- VxLAN (Virtual Extensible LAN) :-

VxLAN creates secure virtual network segments across cloud infrastructure.

Benefits :-

- Network isolation between ~~depap~~ departments.
- Secure communication b/w virtual machines.
- Reduces lateral movement of attackers.

- IDS (Intrusion Detection System) :-

IDS monitors network traffic detects suspicious activities.

Types :-

- Network IDS (NIDS)
- Host IDS (HIDS)

Benefits :-

- Detects malware
- Identifies unauthorized access.
- Alerts administrators in real time.

Secure Architecture :-

- Firewall
- SDN Controller
- VXLAN Segmentation
- IDS/IPS
- VPN for secure remote access
- IAM with MFA
- Backup & disaster Recovery
- SIEM for centralized monitoring.